Informatik I

Woche 2: Ganzzahldivision & Modulo, logische Ausdrücke, Ausdrücke mit gemischten Datentypen

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The Practical Use of The Modulo Operator

Which question is answered by the output of the following piece of code?

```
int a;
std::cin >> a;
if (a % 2 == 0) {
  std::cout << "Yes" << std::endl;
} else {
  std::cout << "No" << std::endl;
}
```

The Practical Use of The Modulo Operator

Which question is answered by the output of the following piece of code?

```
int a;
std::cin >> a;
if (a % 2 == 0) {
  std::cout << "Yes" << std::endl;
} else {
  std::cout << "No" << std::endl;
}
```

Solution: The output answers whether the user input is an even number.

Informatik - Exercise Session
Evaluating Mixed Type Expressions

Evaluating mixed type expressions

1. Which of the following character sequences are not C++ expressions, and why not? Here, `x` and `y` are variables of type `int`.
   a) `(y++ < 0 && y < 0) + 2.0`
   b) `y = (x++ = 3)`
   c) `3.0 + 3 - 4 + 5`
   d) `5 % 4 * 3.0 + true * x++`
2. For all of the valid expressions that you have identified in 1, decide whether these are lvalues or rvalues, and explain your decisions.
3. Determine the values of the expressions and explain how these values are obtained. Assume that initially `x == 1` and `y == -1`.

Evaluating mixed type expressions: Solution 1)

```
(y++ < 0 && y < 0) + 2.0
```

Evaluating mixed type expressions: Solution 1)

```
(y++ < 0 && y < 0) + 2.0
```

```
(-1 < 0 && y < 0) + 2.0  // after this step: y==0
```

Evaluating mixed type expressions: Solution 1)

```
(y++ < 0 && y < 0) + 2.0
```

```
(-1 < 0 && y < 0) + 2.0  // after this step: y==0
(true && y < 0) + 2.0
```

Evaluating mixed type expressions: Solution 1)

```
(y++ < 0 && y < 0) + 2.0
```

```
(-1 < 0 && y < 0) + 2.0  // after this step: y==0
(true && y < 0) + 2.0
(true && false) + 2.0
```

Evaluating mixed type expressions: Solution 1)

```
(y++ < 0 && y < 0) + 2.0
```

```
(-1 < 0 && y < 0) + 2.0  // after this step: y==0
(true && y < 0) + 2.0
(true && false) + 2.0
(false) + 2.0
```

Evaluating mixed type expressions: Solution 1)

```
(y++ < 0 && y < 0) + 2.0
```

```
(-1 < 0 && y < 0) + 2.0  // after this step: y==0
(true && y < 0) + 2.0
(true && false) + 2.0
(false) + 2.0
0.0 + 2.0
```

Evaluating mixed type expressions: Solution 1)

```
(y++ < 0 && y < 0) + 2.0
```

```
(-1 < 0 && y < 0) + 2.0  // after this step: y==0
(true && y < 0) + 2.0
(true && false) + 2.0
(false) + 2.0
0.0 + 2.0
2.0
```

Evaluating mixed type expressions: Solution 1)

```
(y++ < 0 && y < 0) + 2.0
```

```
(-1 < 0 && y < 0) + 2.0  // after this step: y==0
(true && y < 0) + 2.0
(true && false) + 2.0
(false) + 2.0
0.0 + 2.0
2.0
```

Evaluating mixed type expressions: Solution 1)

```
(y++ < 0 && y < 0) + 2.0
```

```
(-1 < 0 && y < 0) + 2.0  // after this step: y==0
(true && y < 0) + 2.0
(true && false) + 2.0
(false) + 2.0
0.0 + 2.0
2.0
```

R-VALUE

Evaluating mixed type expressions: Solution 2)

```
y = (x++ = 3)
```

Evaluating mixed type expressions: Solution 2)

```
y = (x++ = 3)
```

INVALID

Evaluating mixed type expressions: Solution 3)

```
3.0 + 3 - 4 + 5
```

Evaluating mixed type expressions: Solution 3)

```
3.0 + 3 - 4 + 5
```

```
((3.0 + 3) - 4) + 5
```

Evaluating mixed type expressions: Solution 3)

```
3.0 + 3 - 4 + 5
```

```
((3.0 + 3) - 4) + 5
((3.0 + 3.0) - 4) + 5
```

Evaluating mixed type expressions: Solution 3)

```
3.0 + 3 - 4 + 5
```

```
((3.0 + 3) - 4) + 5
((3.0 + 3.0) - 4) + 5
(6.0 - 4) + 5
```

Evaluating mixed type expressions: Solution 3)

```
3.0 + 3 - 4 + 5
```

```
((3.0 + 3) - 4) + 5
((3.0 + 3.0) - 4) + 5
(6.0 - 4) + 5
(6.0 - 4.0) + 5
```

Evaluating mixed type expressions: Solution 3)

```
3.0 + 3 - 4 + 5
```

```
((3.0 + 3) - 4) + 5
((3.0 + 3.0) - 4) + 5
(6.0 - 4) + 5
(6.0 - 4.0) + 5
2.0 + 5
```

Evaluating mixed type expressions: Solution 3)

```
3.0 + 3 - 4 + 5
```

```
((3.0 + 3) - 4) + 5
((3.0 + 3.0) - 4) + 5
(6.0 - 4) + 5
(6.0 - 4.0) + 5
2.0 + 5
2.0 + 5.0
```

Evaluating mixed type expressions: Solution 3)

```
3.0 + 3 - 4 + 5
```

```
((3.0 + 3) - 4) + 5
((3.0 + 3.0) - 4) + 5
(6.0 - 4) + 5
(6.0 - 4.0) + 5
2.0 + 5
2.0 + 5.0
7.0
```

Evaluating mixed type expressions: Solution 3)

```
3.0 + 3 - 4 + 5
```

```
((3.0 + 3) - 4) + 5
((3.0 + 3.0) - 4) + 5
(6.0 - 4) + 5
(6.0 - 4.0) + 5
2.0 + 5
2.0 + 5.0
7.0
```

R-VALUE

Evaluating mixed type expressions: Solution 4)

```
5 % 4 * 3.0 + true * x++
```

Evaluating mixed type expressions: Solution 4)

```
5 % 4 * 3.0 + true * x++
```

```
((5 % 4) * 3.0) + (true * (x++))
```

Evaluating mixed type expressions: Solution 4)

```
5 % 4 * 3.0 + true * x++
```

```
((5 % 4) * 3.0) + (true * (x++))
(1 * 3.0) + (true * (x++))
```

Evaluating mixed type expressions: Solution 4)

```
5 % 4 * 3.0 + true * x++
```

```
((5 % 4) * 3.0) + (true * (x++))
(1 * 3.0) + (true * (x++))
(1.0 * 3.0) + (true * (x++))
```

Evaluating mixed type expressions: Solution 4)

```
5 % 4 * 3.0 + true * x++
```

```
((5 % 4) * 3.0) + (true * (x++))
(1 * 3.0) + (true * (x++))
(1.0 * 3.0) + (true * (x++))
3.0 + (true * (x++))
```

Evaluating mixed type expressions: Solution 4)

```
5 % 4 * 3.0 + true * x++
```

```
((5 % 4) * 3.0) + (true * (x++))
(1 * 3.0) + (true * (x++))
(1.0 * 3.0) + (true * (x++))
3.0 + (true * (x++))
3.0 + (true * 1)
```

Evaluating mixed type expressions: Solution 4)

```
5 % 4 * 3.0 + true * x++
```

```
((5 % 4) * 3.0) + (true * (x++))
(1 * 3.0) + (true * (x++))
(1.0 * 3.0) + (true * (x++))
3.0 + (true * (x++))
3.0 + (true * 1)
3.0 + (1 * 1)
```

Evaluating mixed type expressions: Solution 4)

```
5 % 4 * 3.0 + true * x++
```

```
((5 % 4) * 3.0) + (true * (x++))
(1 * 3.0) + (true * (x++))
(1.0 * 3.0) + (true * (x++))
3.0 + (true * (x++))
3.0 + (true * 1)
3.0 + (1 * 1)
3.0 + 1
```

Evaluating mixed type expressions: Solution 4)

```
5 % 4 * 3.0 + true * x++
```

```
((5 % 4) * 3.0) + (true * (x++))
(1 * 3.0) + (true * (x++))
(1.0 * 3.0) + (true * (x++))
3.0 + (true * (x++))
3.0 + (true * 1)
3.0 + (1 * 1)
3.0 + 1
3.0 + 1.0
```

Evaluating mixed type expressions: Solution 4)

```
5 % 4 * 3.0 + true * x++
```

```
((5 % 4) * 3.0) + (true * (x++))
(1 * 3.0) + (true * (x++))
(1.0 * 3.0) + (true * (x++))
3.0 + (true * (x++))
3.0 + (true * 1)
3.0 + (1 * 1)
3.0 + 1
3.0 + 1.0
4.0
```

Evaluating mixed type expressions: Solution 4)

```
5 % 4 * 3.0 + true * x++
```

```
((5 % 4) * 3.0) + (true * (x++))
(1 * 3.0) + (true * (x++))
(1.0 * 3.0) + (true * (x++))
3.0 + (true * (x++))
3.0 + (true * 1)
3.0 + (1 * 1)
3.0 + 1
3.0 + 1.0
4.0
```

R-VALUE